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Extracted Text

Of the Com Slesa, Misca Madera, were alo fujected to chemical examination, by the fame philosopher. When needed tested best in a crochets, they emitted a perpetraphie fnell of them or feathers, and by further examination, by foliation, the proportion of carbonic matter degraded appeared to be greater, and the proportion of carbonate time lobe, than what was indicated by the refut of the experiments on pencilhenures fields. When natural shells belonging to the fame species, were amended in nitric acid very much diluted, the epidermis separated, and the whole of the carbonate of lime we differed. A gelatinoid diffusence, yearly in a liquid that, remains, ed, but it did not retain the figure of the shell, and ex-philic no appearance at Alorsra Hardure. These fields, therefore, contains a larger portion of gelatinous matter than the porcellaneous fluffs, but the other com-parent part counts entirely of carbonate time.

_Health composed of nuclear-departShells of this description were subjected to familiar experiments with this former. When the common oyster was expelled to a red heat, the effects were the fans as those which were produced by the fame processis on the facies of patches from Madeira. The foliation of the hallmark shell was also funlier, except only little the gelatinous parts are of a greater confidence. When the river, muffel was burnt a crushing, the melted much make, with a frongu find all burnt born or cartilage; the shell become of a dark gray colour, and exfoliated. By foliation in the acids, the proportion of carbonic matter Separated was greatest, and but of carbonaceous oil line obtained was left, than from the other shells on which experiments were made.

My Mother of peat.

When an unburnt fluef of this description was im-mected in diluted nitric oxide, a rapid foliation and effir-vicosecond took place, and at the end of two days the whole of the carbonate of lime was nearly diffused. A feries of means now only remnants, of which the epidemics constituted the full. These membranes fill retained the figure of the shell. The carbonate of lime was a flat readily diffused, because the acid came easily to contain with it; but the proceeds became flower, as it was more difficult for the acid to indicate itself between the different membranes of which the shell is compelled. The halted iris, and the turbo olearies, were found to rethable this muffel, except that the membranaceous parts were more compact and dendric.

When these fields are deprived by an acid of the carbonate of lime, which gives them their birders, they appear to be formed: of different membranes, ap- pled fratum fuer fratum. Each membranes is fur- milled with a corresponding coarse cruml of carbonate of lime, and it is foisted, that it is always between every two membranes, beginning with the epidermis, and ending with the internal membrane, which has been latformed. The animals which inhabit both fleaified flesh, precede their habitation by the addi-tion of a fratum of carbonate of lime, which is fe-cured by a new membrane. And as every additional fratum exceeds in extent that which was previously formed, the shell becomes stronger in properties as it is enlarged; and thus the growth and age of the animal may be denoted by the number of frata of which the thell is compold. Similar experiments were made

C O N C H O L G Y.

Chap 1

on pieces of mother of pearl as they are imported from China, China, and with precisely the fame refute. They ap- peated to be compelled of the fame grafiations matter such as carbonate of lime. In all the thiefs of this description which were named in aside, the membranes parts retained the each figure of the field, and they appeared different to be compelled of fibres, arranged in a parallel direction, correponding to the configuration of the flank.

Pearl–The constituent parts of pearl appear to be Pearl familiar to thief of mother-of-pent. They are compelled concerned costs of membranes and colonies of lines, and resemble in firstature the globular, calcareous con-cretions which are known by the name of _phylidae._ The influence and produced appearance of pearl and another-of-past, evidently depend on their lamellated fracture and femitran/parency.

From their experiments, it appears, that shells are complex composed of surface of lime and gluten. In formas as pure in the porcellaneous shells, the proportion of carbonate. Time is great, while that of the animal matter is finally, and the mean one regarded as the beginning of the feries; while shells that come under the definition of mother-of-post are to be placed into the extra-mity, having a smaller proportion of carbonate of lime, and a greater proportion of membranous fusblance. It is the first the carbonate of lime is merely consumed by the animal matter; in the better the carbonate of lime-ferives to harden the membranous fusblance. But between fields are enzymes, in the proportion of carbone-nate of lime and animal glutes, of which alltaculous fulblances are compelled, there are no doubt numerous intermediates, arising from the nature of the animal to which they from a covering, its peculiar ha-bits, or mode of life.

Sect. II. _Of the Formation of Shells._

Tuz thehell or covering of telacous animals, has been considered as in fome marine analogues to the bones of other animals, although it formation and growth are very different, since it feres as a hole coupport to the mulches, which are attached to its inter-fortified. The principal side of the shell, however, is to force as a covering or defense to the animal.

Telacous animals are not only extremely different in external form, had alls in the mode of their problem. Limtion. Some are viviparous, as the motif of their which people inhabit bivalve shells, multivalves, and even forms of the unitvives, while the others, which form the fire greetate proportions are ostropus. In non point, how ever, they all agree, that whatever be the mode of probilution, whether from an egg, or directly from the uterus of the mother, the fluid is formed on the body of the young animal, and is propontioned to its bulk.

The bell defenders who have yet been made, and made the most elaborate investigation which has hitherto ophawvel peace, concerning the formation and development of iron.

balls, are holes of the colonized Heamur, which were published in the Memoirs of the Academy of Sciences for the year 1790. The fate should be beenpecucul- and by other authors, but their orbits have been mainly the arms so thole of this distinguished naturalidh. Klein is almost the only author who has advanced a different opinion. In his dissertation concerning the formation

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Of the Conſtituent Parts of Shells, &c.

Some ſpecies of patella, which were brought from Madeira, were alſo ſubjected to chemical examination, by the ſame philoſopher. When expoſed to a red heat in a crucible, they emitted a perceptible ſmell of horn or feathers; and by farther examination, by ſolution, the proportion of carbonic matter depoſited appeared to be greater, and the proportion of carbonate of lime leſs, than what was indicated by the reſult of the experiments on porcellaneous ſhells. When unburnt ſhells belonging to the ſame ſpecies, were immerſed in nitric acid very much diluted, the epidermis ſeparated, and the whole of the carbonate of lime was diſſolved. A gelatinous ſubſtance, nearly in a liquid ſtate, remained, but it did not retain the figure of the ſhell, and exhibited no appearance of a fibrous ſtructure. Theſe ſhells, therefore, contain a larger portion of gelatinous matter than the porcellaneous ſhells, but the other component part conſiſts entirely of carbonate of lime.

69 Mother-of-pearl.

Shells compoſed of mother-of-pearl.—Shells of this deſcription were ſubjected to ſimilar experiments with the former. When the common oyſter was expoſed to a red heat, the effects were the ſame as thoſe which were produced by the ſame proceſs on the ſpecies of patella from Madeira. The ſolution of the unburnt ſhell was alſo ſimilar, excepting only that the gelatinous part was of a greater conſiſtency. When the river muſſel was burnt in a crucible, it emitted much ſmoke, with a ſtrong ſmell of burnt horn or cartilage; the ſhell became of a dark gray colour, and exfoliated. By ſolution in the acids, the proportion of carbonic matter ſeparated was greater, and that of carbonate of lime obtained was leſs, than from the other ſhells on which experiments were made.

When an unburnt ſhell of this deſcription was immerſed in diluted nitric acid, a rapid ſolution and effervescence took place; and at the end of two days, the whole of the carbonate of lime was nearly diſſolved. A ſeries of membranes now only remained, of which the epidermis conſtituted the firſt. Theſe membranes ſtill retained the figure of the ſhell. The carbonate of lime was at firſt readily diſſolved, becauſe the acid came eaſily in contact with it; but the proceſs became ſlower, as it was more difficult for the acid to inſinuate itſelf between the different membranes of which the ſhell is compoſed. The haliotis iris, and the turbo olearius, were found to reſemble this muſſel, except that the membranaceous parts were more compact and denſe.

When theſe ſhells are deprived by an acid of the carbonate of lime, which gives them their hardneſs, they appear to be formed of different membranes, applied ſtratum ſuper ſtratum. Each membrane is furniſhed with a correſponding coat or cruſt of carbonate of lime, and it is ſo ſituated, that it is always between every two membranes, beginning with the epidermis, and ending with the internal membrane, which has been laſt formed. The animals which inhabit theſe ſtratified ſhells, increaſe their habitation by the addition of a ſtratum of carbonate of lime, which is ſecured by a new membrane. And as every additional ſtratum exceeds in extent that which was previouſly formed, the ſhell becomes ſtronger in proportion as it is enlarged; and thus the growth and age of the animal may be denoted by the number of ſtrata of which the ſhell is compoſed. Similar experiments were made on pieces of mother-of-pearl as they are imported from China, and with preciſely the ſame reſults. They appeared to be compoſed of the ſame gelatinous matter and carbonate of lime. In all the ſhells of this deſcription which were immerſed in acids, the membranaceous parts retained the exact figure of the ſhell, and they appeared diſtinctly to be compoſed of fibres, arranged in a parallel direction, correſponding to the configuration of the ſhell.

70 Pearl.

Pearl.—The conſtituent parts of pearl appear to be ſimilar to thoſe of mother-of-pearl. They are compoſed of concentric coats of membrane and carbonate of lime, and reſemble in ſtructure the globular, calcareous concretions which are known by the name of *oſſolithes*. The irideſcence and undulated appearance of pearl and mother-of-pearl, evidently depend on their lamellated ſtructure and ſemitranſparency.

71 Compoſition of parts.

From theſe experiments it appears, that ſhells are compoſed of carbonate of lime and gluten. In ſome, as in the porcellaneous ſhells, the proportion of carbonate of lime is great, while that of the animal matter is ſmall; and theſe may be regarded as the beginning of the ſeries; while ſhells that come under the deſcription of mother-of-pearl are to be placed at the other extremity, having a ſmaller proportion of carbonate of lime, and a greater proportion of membranaceous ſubſtance. In the firſt the carbonate of lime is merely cemented by the animal matter; in the latter the carbonate of lime ſerves to harden the membranaceous ſubſtance. But between theſe two extremes, in the proportion of carbonate of lime and animal gluten, of which all teſtaceous ſubſtances are compoſed, there are no doubt numerous intermediate gradations, ariſing from the nature of the animal to which they form a covering, its peculiar habits, or mode of life.

Sect. II. *Of the Formation of Shells.*

The ſhell or covering of teſtaceous animals, has been conſidered as in ſome meaſure analogous to the bones of other animals, although its formation and growth are very different, ſince it ſerves as a baſe or ſupport to the muſcles, which are attached to its internal ſurface. The principal uſe of the ſhell, however, is to ſerve as a covering or defence to the animal.

72 Teſtaceous animals oviparous and viviparous.

Teſtaceous animals are not only extremely different in external form, but alſo in the mode of their production. Some are viviparous, as the moſt of thoſe which inhabit bivalve ſhells, multivalves, and even ſome of the univalves; while the others, which form the far greater proportion, are oviparous. In one point, however, they all agree, that whatever be the mode of production, whether from an egg, or directly from the uterus of the mother, the ſhell is formed on the body of the young animal, and is proportioned to its bulk.

73 Reaumur's inveſtigation.

The beſt obſervations which have yet been made, and the moſt elaborate inveſtigation which has hitherto appeared, concerning the formation and developement of ſhells, are thoſe of the celebrated Reaumur, which were publiſhed in the Memoirs of the Academy of Sciences for the year 1709. The ſame ſubject has been proſecuted by other authors, but their reſults have been nearly the ſame as thoſe of this diſtinguiſhed naturaliſt. Klein is almoſt the only author who has advanced a different opinion. In his diſſertation concerning the formation of

Figure 1: *
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